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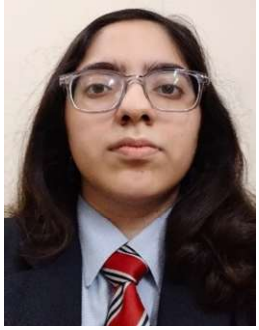
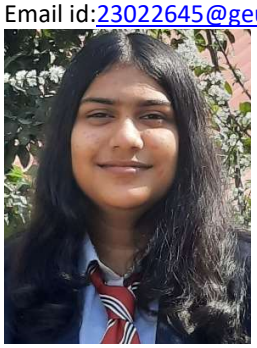
PROJECT AND TEAM INFORMATION

Project Title

Disaster Response and Resource Management System

Student/Team Information

Team Name:	Trinity Tech
Ria Kataria (Team Lead)	Name: Ria Kataria Student id: 23022107 Email: 23022107@geu.ac.in 
Madhavi Payal	Name: Madhavi Payal Student id: 23022085 Email id: 23022085@geu.ac.in

Sanmita Paul	Name: Sanmita Paul Student id: 23022645 Email id:23022645@geu.ac.in 

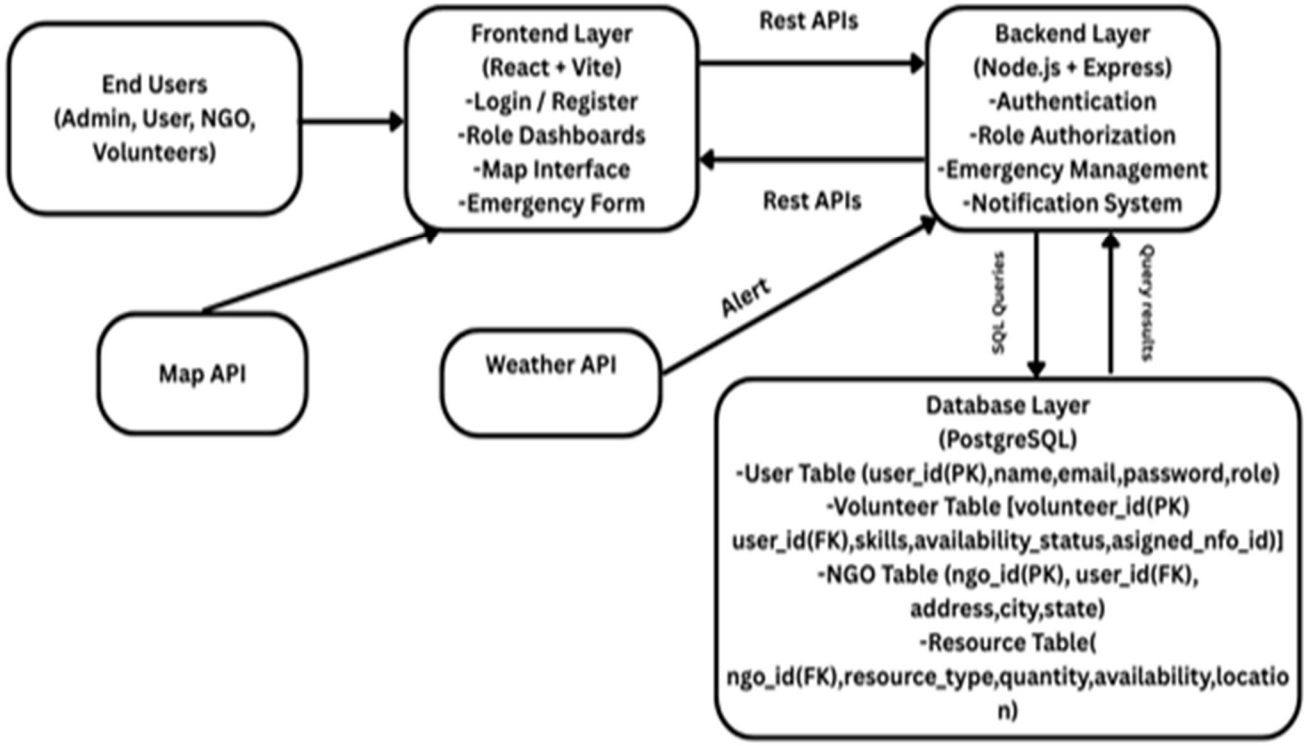
PROJECT PROGRESS DESCRIPTION

Project Abstract

This project aims to create a Disaster Management System that will help in the efficient coordination of users, administrators, NGOs, and volunteers in the event of a disaster. The proposed system will allow users to register, log in, and access the different roles available. This will help in the efficient response time and better management of disaster information through a centralized database and web interface.

Updated Project Approach and Architecture

- The project uses a full-stack approach to web development. The front-end is implemented by React (TypeScript) for developing responsive UI components like the login and signup pages. The back-end is implemented by Node.js/Express for handling API routes like /signup and /login.
- The project uses a cloud database like PostgreSQL (Supabase) for storing data. The communication between the front-end and the back-end is achieved by REST API calls, and the protocol used is HTTP (fetch). The passwords are stored by bcrypt hashing.



Tasks Completed

Task Completed	Team Member
Signup page UI	Sanmita Paul
Login page UI	Sanmita Paul
Backend setup	Madhavi Payal
PostgreSQL database setup (Supabase)	Madhavi Payal
Signup API integration	Madhavi Payal
Login API with bcrypt authentication	Madhavi Payal
Database connection using pooler	Ria Kataria
Fixing connection errors & DNS issues	Ria Kataria

Challenges/Roadblocks

- The team had a few challenges while developing the application. First, there was an issue with database connectivity, which was causing an ENOTFOUND error. This was mainly attributed to the incompatibility of IPv4 and IPv6 with Supabase's direct connection. This problem was solved by using the Supabase session pooler. Network problems on the laptops of the team members were also solved by configuring them properly.

- Another problem experienced was related to environment variables. For instance, the .env file was pushed to GitHub by mistake. This problem was solved by deleting it and adding it to .gitignore. Password security was also improved by using bcrypt instead of plain text.

Tasks Pending

Task Pending	Team Member (to complete the task)
Role-based Dashboard(UI)	Sanmita Paul
Role-based Dashboard(Backend)	Ria Kataria
Disaster reporting module	Ria Kataria
Resource allocation system	Madhavi Payal
UI improvements & styling	Sanmita Paul
Deployment	Ria Kataria and Madhavi Payal
Testing	Sanmita Paul

Project Outcome/Deliverables

The final outcome will be a fully functional web-based application of the Disaster Management System. It will have user authentication, roles, and database integration. It will be efficient in handling data and communication among the stakeholders. The application will be deployed along with the source code.

Progress Overview

The project is currently 40% complete. The core functional modules, such as authentication, database integration, and backend API, are already done. Some UI enhancements and additional features are still to be completed. The team is a little behind on the feature development side, but they are on track to deliver the critical modules

Codebase Information

- Repository: https://github.com/Sanmita-Paul/Disaster_Management.git
- Branch: main
- Important commits:
  - Backend setup and API creation
  - Supabase integration
  - bcrypt password hashing implementation
  - .env configuration

Testing and Validation Status

Test Type	Status (Pass/Fail)	Notes
Signup API	Pass	User stored in DB
Login API	Pass	Password matched using bcrypt
Database Connection	Pass	Using Supabase pooler
UI Testing	In Progress	Needs improvements

Deliverables Progress

- Authentication System → Completed
- Database Integration → Completed
- UI (Login/Signup) → Completed
- Role-based Features → In Progress
- Deployment → Pending